Boundary-condition geometry for energy-minimal Schwinger pair creation in coherent electromagnetic fields

J. Alexis Román Martínez[1]

[1] *Independent Researcher, Platino Labs*

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Direct pair creation from the quantum vacuum requires electromagnetic fields comparable to the Sauter–Schwinger critical field $E_{\rm cr} = m_e^2 c^3/(e\hbar) \simeq 1.32\times10^{18}\,\mathrm{V\,m^{-1}}$, five to six orders of magnitude above present laser intensities. Because the local pair-production rate is a function of the two Lorentz invariants of the field, the only controllable levers at fixed total energy are the geometry and the coherence of the field superposition. We cast the problem as a variational one: minimise the total input pulse energy subject to a prescribed pair yield. Three statements are established analytically. (i) A single plane wave is a null field, $\mathcal{F} = \mathcal{G} = 0$, and creates no pairs; pair creation is therefore a property of the *collision* of fields, not of an individual pulse. (ii) For N phase-matched beams focused into a common volume at fixed total energy, the peak field scales as $\sqrt{N}$, so the energy required to reach a given invariant scales as $1/N$, under explicitly stated assumptions. (iii) The residual freedom is polarisation and temporal structure, which decide whether the strong-field region is oscillatory (counter-propagating linear standing wave) or quiescent (co-rotating circular collision); the quiescent configuration is preferable for adiabatic pair creation. We state the boundary conditions implied by phase matching and propose a falsifiable one- and two-dimensional particle-in-cell test.

I. INTRODUCTION

The vacuum of quantum electrodynamics is not an empty stage. It is a polarisable medium, and it becomes unstable when the applied electric field approaches

$$E_{\rm cr} = \frac{m_e^2 c^3}{e\hbar} \simeq 1.32\times10^{18}\ \mathrm{V\,m^{-1}}, \tag{1}$$

the field defined in Eq. (1), for which the work performed over one reduced Compton wavelength equals the rest energy of an electron–positron pair [1, 2]. At and beyond $E_{\rm cr}$ the tunnelling rate out of the vacuum is non-perturbative, and a static electric field decays by spontaneously emitting pairs [3]. The corresponding focal intensity, of order $10^{29}\,\mathrm{W\,cm^{-2}}$, is five to six orders of magnitude above what present-day laser facilities deliver (10^{22}–$10^{23}\,\mathrm{W\,cm^{-2}}$) [4].

Three routes are pursued to close that gap. The first is brute force: more energy on target. The second is *dynamically assisted* Schwinger production, in which a high-energy seed particle or photon supplies the energy while the field supplies the non-perturbative character of the process [5]; its beam–beam variant, in which two real photons annihilate into a pair, is Breit–Wheeler production [6]. The third — the subject of this work — is *geometry and coherence*: to arrange the electromagnetic field so that a target invariant is reached with the smallest possible total pulse energy.

This last route rests on a structural fact. The local pair-production rate in a slowly varying field depends on the field only through its two Lorentz invariants [2],

$$\mathcal{F} = \tfrac{1}{2}\left(\mathbf{E}^2 - \mathbf{B}^2\right), \qquad \mathcal{G} = \mathbf{E}\cdot\mathbf{B} \qquad (\hbar = c = 1), \tag{2}$$

or, equivalently, through the two rest-frame field magnitudes a, b they fix (App. A). A single plane wave satisfies $|\mathbf{E}| = |\mathbf{B}|$ and $\mathbf{E} \perp \mathbf{B}$, hence $\mathcal{F} = \mathcal{G} = 0$: it is a *null field*, and it cannot create pairs however intense it becomes. Pair creation is therefore not a property of a pulse but of a *superposition* of non-collinear fields. Geometry is not a refinement of the problem; it is its mechanism.

The remainder of this paper is organised as follows. Sec. II fixes the framework: invariants, rest-frame fields, the effective action, and the variational statement of the optimisation. Sec. III derives the analytic nucleus: the null-field lemma, the invariants of a two-beam collision in the linear and circular configurations, the $1/N$ energy scaling of coherent N-beam focusing, and the boundary conditions imposed by phase matching. Sec. IV states what would falsify the scaling, and outlines the numerical test. Throughout, units are natural ($\hbar = c = 1$, Heaviside–Lorentz) unless a SI value is explicitly written; the metric signature is $(+,-,-,-)$.

II. ANALYTICAL FRAMEWORK

A. Field tensor, invariants and rest-frame fields

Let $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ be the field-strength tensor and ${}^\star F^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ its dual. The two independent Lorentz scalars are

$$\begin{aligned} \mathcal{F} &= -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}\left(\mathbf{E}^2 - \mathbf{B}^2\right), \\ \mathcal{G} &= -\tfrac{1}{4}F_{\mu\nu}{}^\star F^{\mu\nu} = \mathbf{E}\cdot\mathbf{B}. \end{aligned} \tag{3}$$

They determine the two eigenvalues of F, i.e. the field magnitudes seen in the frame in which the field is maximally "electric" or maximally "magnetic". Defining

$$a = \sqrt{\sqrt{\mathcal{F}^2 + \mathcal{G}^2} + \mathcal{F}}, \qquad b = \sqrt{\sqrt{\mathcal{F}^2 + \mathcal{G}^2} - \mathcal{F}}, \tag{4}$$

one has

$$a^2 - b^2 = 2\mathcal{F}, \qquad a^2 b^2 = \mathcal{G}^2, \qquad a^2 + b^2 = 2\sqrt{\mathcal{F}^2 + \mathcal{G}^2}. \tag{5}$$

There always exists a Lorentz frame in which $\mathbf{E} \parallel \mathbf{B}$, with $|\mathbf{E}| = a$, $|\mathbf{B}| = b$. In particular, when $\mathcal{G} = 0$ and $\mathcal{F} > 0$ the field is purely electric in some frame, with $|\mathbf{E}| = a = \sqrt{2\mathcal{F}}$ and $\mathbf{B} = 0$. The quantity a is the field that governs the tunnelling rate, and it — not the lab intensity — is the object the geometry must maximise. A derivation is given in App. A.

B. Effective action and the pair-creation yield

The vacuum persists as a polarisable medium through the one-loop correction to the Maxwell action. To fourth order in the field the Heisenberg–Euler Lagrangian reads [3]

$$\mathcal{L}_{\mathrm{EH}} = \frac{2\alpha^2}{45\, m_e^4}\left[\left(\mathbf{E}^2 - \mathbf{B}^2\right)^2 + 7\left(\mathbf{E}\cdot\mathbf{B}\right)^2\right], \tag{6}$$

The imaginary part of Eq. (6), continued to the appropriate configuration, gives the tunnelling rate out of the vacuum [2]. For a uniform electric field E the rate per unit volume and time is

$$w(E) = \frac{(eE)^2}{4\pi^3}\sum_{n=1}^{\infty}\frac{1}{n^2}\exp\left(-\frac{n\pi m_e^2}{eE}\right) \qquad (\hbar = c = 1), \tag{7}$$

whose leading term is exponentially suppressed by E_{cr}/E. For a slowly varying field the rate generalises through the *locally constant field approximation* (LCFA): the local rate is $w = w(a(\mathbf{x}, t))$, a function of the rest-frame field a of Eq. (4) evaluated pointwise. The number of pairs produced by a spacetime field configuration is then the yield functional

$$\mathcal{N}[\{A_i^\mu\}] = \int_\Omega d^4x\; w(a(\mathbf{x}, t))\,, \qquad a = a(\mathcal{F}, \mathcal{G})\,, \tag{8}$$

with the invariants $\mathcal{F}, \mathcal{G}$ of Eq. (2) built from $F = \sum_i F_i(A_i)$ by the superposition of the individual beam potentials A_i^μ (valid in vacuum, below the pair-creation threshold).

C. The variational problem

Let N beams carry individual energies $\mathcal{E}_i$, and let $\mathcal{E}_{\mathrm{in}} = \sum_i \mathcal{E}_i$ be the total input energy. We pose the optimisation as

$$\boxed{\begin{array}{l}\displaystyle\min_{\{A_i^\mu\}}\ \mathcal{E}_{\mathrm{in}} = \sum_{i=1}^{N}\mathcal{E}_i, \quad \text{subject to} \\ \mathcal{N}[\{A_i^\mu\}] \ \geq\ \mathcal{N}_*,\ \partial_\mu F^{\mu\nu} = 0\end{array}} \tag{9}$$

where $F = \sum_i F_i(A_i)$, and each A_i^μ is subject to the physical realisability constraints of the source (maximum focal intensity, pulse duration, and focal spot size). The geometry — crossing angles, relative phases, polarisations and focal offsets — is encoded entirely in the $\{A_i^\mu\}$; the pair-yield target $\mathcal{N}_*$ is fixed by the observational goal (single pair, a detectable signal, a cascade). Equation (9) is the precise statement of "reach the threshold with the least energy on target".

III. PULSE GEOMETRY AND BOUNDARY CONDITIONS

A. A single plane wave is null (Lemma 1)

For a plane wave with wave vector $\mathbf{k}$ and unit vector $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$, $\mathbf{B} = \hat{\mathbf{k}}\times\mathbf{E}$ and $|\mathbf{E}| = |\mathbf{B}|$. Hence $\mathcal{G} = \mathbf{E}\cdot\mathbf{B} = 0$ and $\mathcal{F} = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) = 0$ identically, at every instant and every phase. By Eq. (8) the yield vanishes: $w(a = 0) = 0$. For a field transverse to a single propagation direction, an arbitrarily intense *single* pulse therefore creates no pairs from the vacuum, and pair creation requires at least two fields whose wave vectors are not parallel. The null is exact for a plane wave and holds for a paraxial beam; it fails at the focus of a tightly focused pulse, where $\nabla\cdot\mathbf{E} = 0$ forces a longitudinal component $E_z \sim E_\perp/(kw_0)$ and the time-averaged invariant acquires a term of relative order $(kw_0)^{-2}$. The results below are stated for transversality-preserving fields.

B. Two counter-propagating beams (Lemma 2)

Consider two plane waves of equal amplitude E_0 and frequency ω, propagating along $\pm\hat{\mathbf{z}}$ and focused into a common volume.

a. (a) Linear, co-polarised. Take both polarised along $\hat{\mathbf{x}}$. Writing $\omega t - kz$ and $\omega t + kz$ for the two phases, the superposition is

$$\mathbf{E} = 2E_0\cos\omega t\,\cos kz\;\hat{\mathbf{x}}, \qquad \mathbf{B} = 2E_0\sin\omega t\,\sin kz\;\hat{\mathbf{y}}. \tag{10}$$

The fields of Eq. (10) yield the invariants

$$\mathcal{F} = E_0^2(\cos 2\omega t + \cos 2kz)\,, \qquad \mathcal{G} = 0, \tag{11}$$

since $\mathbf{E} \parallel \hat{\mathbf{x}}$ and $\mathbf{B} \parallel \hat{\mathbf{y}}$ are everywhere orthogonal. This is an oscillatory standing wave: $\mathcal{F}$ oscillates in time and changes sign. At its spatio-temporal antinodes, $\cos\omega t\cos kz = \pm 1$, one has $|\mathbf{E}| = 2E_0$, $\mathbf{B} = 0$, and $\mathcal{F} = 2E_0^2$: a pure-electric region at twice the single-beam amplitude. The field, however, passes through zero at the same location, and with it the yield.

b. (b) Circular, co-rotating. Take both waves circularly polarised with the same helicity. Then

$$\begin{aligned}\mathbf{E} &= 2E_0\cos kz\,(\cos\omega t,\ \sin\omega t,\ 0),\\ \mathbf{B} &= 2E_0\sin kz\,(\cos\omega t,\ \sin\omega t,\ 0),\end{aligned} \tag{12}$$

so that

$$\mathcal{F} = 2E_0^2 \cos 2kz, \qquad \mathcal{G} = 2E_0^2 \sin 2kz, \tag{13}$$

both independent of time. In Eq. (12) the fields $\mathbf{E}$ and $\mathbf{B}$ are parallel, so $\mathcal{G}$ vanishes only on the planes $kz = n\pi/2$; the strong-field planes $kz = n\pi$ are pure electric, with $\mathcal{F} = 2E_0^2$, $|\mathbf{E}| = 2E_0$ and $\mathbf{B} = 0$, and they persist. Moreover $\mathcal{F}^2 + \mathcal{G}^2 = 4E_0^4$ everywhere, so the total rest-frame magnitude $\sqrt{a^2 + b^2} = 2E_0$ is spatially uniform, the configuration rotating from purely electric at $kz = n\pi$ to purely magnetic at $kz = (n + \frac{1}{2})\pi$. Where the linear configuration gives the same peak but a field that crosses zero twice per period, the co-rotating circular configuration gives a *quiescent* strong field that is stationary in time. Since the tunnelling rate (7) depends on the field magnitude and the adiabaticity ratio $\omega/(eE)$, a strong-field region that does not cross zero in time is preferable for generating pairs from the vacuum. The two temporal profiles are compared in Fig. 1(a). This is a discrete geometric boundary condition on the polarisation, stated in Sec. III D.

C. Coherent N-beam focusing reduces the required energy (Proposition)

Let N beams of equal amplitude E_b be phase-matched at a common focus of fixed area Σ and duration τ, so that their fields add coherently: $E_{\text{peak}} = NE_b$. For a beam of fixed Σ, τ the beam energy is $\mathcal{E}_b = \kappa E_b^2$ with a constant κ depending only on Σ and τ. To reach a target peak field E_*,

$$\begin{aligned} E_b = \frac{E_*}{N} \quad \Longrightarrow \quad \mathcal{E}_{\text{in}} &= N\mathcal{E}_b = N\kappa\left(\frac{E_*}{N}\right)^2 \\ &= \frac{\kappa E_*^2}{N} = \frac{\mathcal{E}_{\text{ref}}}{N}, \end{aligned} \tag{14}$$

where $\mathcal{E}_{\text{ref}} = \kappa E_*^2$ is the energy required with a single beam. Equivalently, at fixed total energy the peak field scales as $\sqrt{N}$ and the intensity as N. The total energy needed to reach a given invariant therefore falls as $1/N$ (Fig. 1(b)). Setting the target field to the critical value, $E_* = E_{\text{cr}}$, the input energy required to reach the pair-creation threshold falls as $1/N$ relative to the threshold of a single beam.

The algebra here is that of coherent addition and is elementary; what the proposition establishes is that the linear-in-N reduction of the threshold energy follows from vacuum superposition alone, with no further assumption about the interaction. The assumptions are explicit and each is falsifiable: (i) vacuum superposition, $F = \sum_i F_i$, valid below the pair-creation threshold; (ii) perfect phase matching, i.e. path-length differences $\Delta L \ll \lambda$ across all beams; (iii) a fixed, common focal volume $\Sigma \times \tau$; (iv) no plasma formation during the rise, which would spoil the vacuum superposition. Practical limits — the finite number of beams a chamber can accept, diffraction, optical damage, and the alignment tolerance quantified below — bound the attainable N and are the subject of Sec. IV.

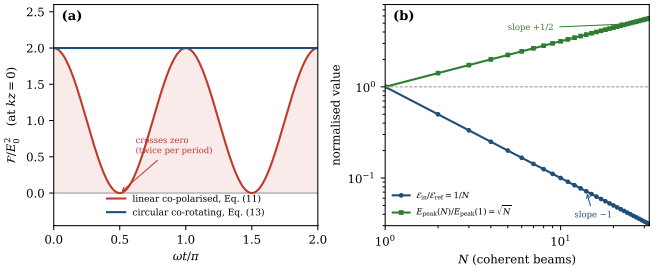


FIG. 1. (a) The invariant $\mathcal{F}$ at the strong-field plane $kz = 0$ as a function of time, for the two polarisation geometries. The counter-propagating linear standing wave, Eq. (11), oscillates through zero twice per period, so the tunnelling rate is switched off for half of the cycle; the co-rotating circular collision, Eq. (13), is stationary at $\mathcal{F} = 2E_0^2$. (b) Energy scaling of N coherently combined beams at the common focus: the input energy required to reach a prescribed invariant falls as $1/N$ (Eq. (14)), while at fixed total energy the peak field grows as $\sqrt{N}$. The dashed line marks the single-beam reference. Both panels are analytic; no simulation input enters.

D. Phase-matching boundary conditions

Equation (9) is an optimisation over fields, but the fields are not free: they must satisfy $\partial_\mu F^{\mu\nu} = 0$ and, to add coherently, must arrive at the focus with a common phase. This translates into geometric boundary conditions on the beam arrangement:

1. *Phase alignment.* The optical path from each source to the common focus must be equal to better than a fraction of a wavelength, $\Delta L \ll \lambda$; otherwise the relative phases drift and E_{peak} falls below NE_b. The tolerance on the crossing angle follows from $\Delta L \simeq d\,\delta\theta$ for beams of transverse extent d.

2. *Polarisation alignment.* For the quiescent (circular) optimum of Eq. (13), all beams must share the *same* helicity; a helicity flip converts the configuration towards the oscillatory linear case of Eq. (11) and destroys the time-independence of $\mathcal{F}$.

3. *Symmetry.* A symmetric arrangement about the focus equalises the ΔL of all beams simultaneously and is the configuration that maximises N for a given alignment tolerance.

These conditions are the concrete content of "boundary-condition geometry": they select, among all superpositions satisfying Maxwell's equations, the subclass that is (a) coherent and (b) quiescent.

IV. DISCUSSION, FALSIFIABILITY AND OUTLOOK

The program sketched above makes a sharp, falsifiable prediction and leaves a well-defined residual problem.

a. What would falsify the $1/N$ scaling. Equation (14) assumes vacuum superposition and perfect coherence. It is refuted if, in a controlled calculation in which N is increased at fixed total energy and fixed Σ, τ, the peak rest-frame field a fails to scale as $\sqrt{N}$ — for instance because the crossing region is driven into plasma, because phase-matching errors grow with N, or because back-reaction from the created pairs screens the field. Each of these is a calculable departure, and each has a distinct signature in the emerging pair spectrum.

b. The residual problem. Once N and the polarisation are fixed, the open question is the optimal *temporal* profile: how the field must be shaped in time so that a is held above the threshold long enough for the rate (7) to accumulate, without the oscillating zero-crossings that suppress the yield. This couples the geometry to the adiabaticity of the process and is where a purely analytic treatment gives way to computation.

c. Numerical test. The analytic statements above are testable at low cost in one and two dimensions with open particle-in-cell codes that implement the quantum processes: SMILEI [7], EPOCH [8], OSIRIS, and WarpX [9]. The measurement to perform is the *rest-frame field* $a(\mathbf{x}, t)$ reconstructed from the simulated $\mathbf{E}, \mathbf{B}$ and the yield obtained by integrating Eq. (8), compared against the analytic predictions (11)–(13) for two beams and the scaling (14) for N beams. The laboratory context for this regime is LUXE at DESY [10], whose beam–laser and photon–laser configurations are coherent superpositions of the same kind — of two beams, not of the optimised N-beam geometry — so that they constrain the strong-field framework and the two-beam invariants (11)–(13), rather than the energy scaling of Eq. (14).

d. Scope. This draft deliberately restricts itself to the pure-field problem. It does not treat dynamically assisted production, and it makes no claim beyond the assumptions stated at Eq. (14). A claim that survives those assumptions is a result; a claim that needs them dropped is a computation.

A. Domain of validity and practical constraints

The bounds collected here are part of the result rather than qualifications attached to it: inside them, the scaling of Eq. (14) is what this paper states; outside them, the same question is answered by a different calculation.

a. Transversality. The paraxial estimate of Sec. III leaves a longitudinal component $E_z \sim E_\perp/(kw_0)$ and a relative correction to the invariant of order $(kw_0)^{-2}$. For a waist of two wavelengths, $kw_0 = 4\pi \approx 12.6$ and that correction is 6.3×10^{-3}; it falls below 10^{-2} for $kw_0 \gtrsim 10$, i.e. for waists wider than about 1.6λ. At an $f/1$ focus $kw_0 \sim 1$ and the same expression is of order unity, so the transverse invariants no longer determine the local field, the yield functional of Eq. (8) ceases to approximate the local rate, and the scaling lies outside the regime in which it is stated. Tight focusing used to reach the critical field with a single pulse is therefore not a more favourable configuration of this scheme but a separate calculation.

b. Adiabaticity and the coherence budget. The rate of Eq. (7) enters in its locally constant form, which requires the field to vary slowly on the tunnelling time. The controlling ratio is the adiabaticity parameter $\gamma \equiv m_e\omega/(eE)$ — the ratio $\omega/(eE)$ of Sec. III written in units of the electron mass — equivalently the inverse of the dimensionless amplitude $a_0 \equiv eE/(m_e\omega)$; neither is the rest-frame field a of Eq. (4). Over the amplitudes reached here, a_0 runs from 1.7×10^2 to 6.8×10^2, so γ runs from 5.9×10^{-3} down to 1.5×10^{-3}: the pair-creation time $m_e c/(eE)$ is shorter than the optical period by a factor $2\pi a_0$, i.e. by three to four orders of magnitude, and the field is locally static to better than one part in 10^2 over the production time. Pulse length in cycles is thus not what limits the validity of Eq. (7), and the margin widens with N, since at fixed total energy $a_0 \propto \sqrt{N}$.

The binding constraint is instead the coherence of the combining. If the N beams arrive with residual phase differences of rms σ_ϕ, the coherent sum of Eq. (14) is partially randomised and the exponent β fitted to the peak field as a function of N falls below $1/2$: the acceptance band $0.45 \le \beta \le 0.55$ is left at $\sigma_\phi^* \simeq 0.55$ rad, about 0.09 of an optical period, which for an 800 nm driver is $\sigma_\phi^* \simeq 0.24$ fs rms. This is a specification on the combining optics, not a defect of the model: it identifies the one quantity to which the result is sensitive, and with fully random phases the gain vanishes and no scaling remains to be tested.

Appendix A: Eigenvalues of the field tensor and the rest-frame fields

The invariants (3) are the coefficients of the characteristic polynomial of $F^{\mu\nu}$. Contracting $F^{\mu}{}_{\nu}F^{\nu}{}_{\lambda}$ and using the two invariants one finds that the non-zero eigenvalues come in a pair $\pm a$ and a pair $\pm ib$ with

$$a^2 = \sqrt{\mathcal{F}^2 + \mathcal{G}^2} + \mathcal{F}, \qquad b^2 = \sqrt{\mathcal{F}^2 + \mathcal{G}^2} - \mathcal{F}, \quad \text{(A1)}$$

which is Eq. (4); the relations (5) follow immediately. Because $\mathcal{F}$ and $\mathcal{G}$ are invariant, so are a and b: they are the field magnitudes in the frame where $\mathbf{E} \parallel \mathbf{B}$ (if $\mathcal{G} \neq 0$) or where one of the two vanishes. For a null field ($\mathcal{F} = \mathcal{G} = 0$) both eigenvalues vanish, recovering Lemma 1. For a purely electric configuration ($\mathcal{G} = 0$, $\mathcal{F} > 0$) one has $b = 0$ and $a = \sqrt{2\mathcal{F}} = |\mathbf{E}|$ in the rest frame, which is why $\mathcal{F}$ alone measures how close a configuration is to the Schwinger threshold.

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